

PSAP-Specific Cybersecurity Awareness Training: Development and Evaluation with a Regional 911 Dispatch Center

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Bachelor of Applied Science in Cybersecurity

Capstone Project

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June 11, 2026

Abstract

Public Safety Answering Points (PSAPs), the 911 centers that form the first and often only link to emergency services, operate under the exact conditions that social engineering is built to exploit: mission-critical systems that cannot be paused, constant time pressure, a professional instinct to help and to defer to authority, and a large public information footprint that makes vendor impersonation convincing. Yet most awareness training delivered to dispatch staff is drawn from enterprise frameworks that do not account for these conditions or the attack vectors most likely to target a 911 center. This capstone developed and delivered a PSAP-specific cybersecurity awareness training program in partnership with a regional 911 public safety answering point and its dispatch center, focused on spear phishing, spear vishing, and vendor impersonation. An anonymous pre-training survey measured both behavioral response to realistic scenarios and self-assessed confidence, and its results directly shaped three deliverables: a training presentation, a facilitator script, and a quick-reference card. A matching post-training survey measured change. The baseline revealed a confident, generally trained cohort that still carried specific behavioral gaps, concentrated in link verification and independent callback verification. After training, the link-verification gap closed completely while the callback-verification gap remained unresolved. Because the instruments were anonymous and the cohort small and single-site, results are reported as directional rather than generalizable. The project contributes both a replicable method for identifying and targeting PSAP-specific gaps and a finding that operationally specific training can close the behavioral gap it targets, though not automatically every gap.

Keywords: PSAP, 911 dispatch, social engineering, spear phishing, vishing, security awareness training, vendor impersonation

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Introduction

Public Safety Answering Points (PSAPs), commonly referred to in the United States as 911 centers, are the immediate and often only link to emergency services when help is needed. PSAPs in the United States handle an estimated 240 million calls each year, roughly 657,000 calls per day, or about 7.6 calls per second (National Emergency Number Association, n.d.). They sustain an ever-increasing volume of calls, third-party digital connections like smart device alerting, automated crash detection, etc., and outside vendor relationships like telecommunications vendors, emergency services technology vendors, and the like. All of those connections help to keep the system working 24 hours a day, every day. Unlike most other organizations, PSAP infrastructure is truly mission-critical: 911 systems cannot be taken offline for maintenance or recovery without considerable planning, or there's direct harm done to public safety. And dispatch staff cannot just pause a call or walk away from their consoles to verify if a suspicious phone call or email is a malicious attempt to compromise the system.

This capstone project developed and delivered a PSAP-specific cybersecurity awareness training program in partnership with a regional 911 public safety answering point in the Pacific Northwest (referred to here as "the agency") and its dispatch center. The project is scoped to address two threat categories identified as most relevant to the PSAP environment: spear phishing and spear vishing, with a specific emphasis on vendor impersonation tactics. These focuses were selected based on documented real-world PSAP incidents and conversations with the agency's training leadership. The training was designed to complement the agency's existing security awareness training, not to replace it. Deliverables produced for the agency include a 15-slide training presentation, a facilitator script, and a physical quick-reference card for staff to retain after the session. An anonymous pre-training survey was distributed and analyzed to

identify knowledge gaps, and a matched post-training survey measured behavioral and attitudinal changes.

Cyberattacks on emergency communications are not theoretical. Loukas et al. (2013) surveyed the cybersecurity threats facing emergency management infrastructure. They noted that these systems occupy a uniquely exposed position because successful attacks carry outsized consequences: a disrupted enterprise loses productivity. In contrast, a disrupted PSAP loses its ability to answer emergency calls and dispatch emergency services. The real-world record confirms this, even locally. In 2018, a CenturyLink network configuration failure cascaded into a multi-state 911 outage, and the Washington State Attorney General's investigation found that it blocked an estimated 10,752 911 calls during the outage, including active emergencies in the Tri-Cities (Washington State Office of the Attorney General, 2021). In 2016, a JavaScript exploit shared on social media forced iPhones to auto-dial 911 repeatedly, flooding a dispatch center in Surprise, Arizona, with more than 100 calls within minutes and putting it in immediate danger of losing operational capability to a telephony denial of service (TDoS) attack (The Hacker News, 2016). Dameff et al. (2014) demonstrated at DEF CON 22 how PSAP infrastructure could be targeted, identifying specific vulnerabilities in call routing, caller ID, and third-party integrations. These are not isolated incidents: PSAPs experience a significant cyber incident on average every 60 days, with recovery times that can extend up to 25 days (Motorola Solutions Public Safety Threat Alliance [PSTA], 2025).

Technical controls address only part of the attack surface. The most persistent attacks do not defeat firewalls; they go around them by targeting people. Williams et al. (2018) found that phishing messages conveying authority significantly increased click rates and that combining authority with urgency and fear further reduced recipients' ability to evaluate messages critically.

Dispatch staff are trained to prioritize urgent requests, help when and where they can, and defer to authority, so an attacker who impersonates a vendor or system administrator and frames a request as urgent is deliberately engineering conditions that the PSAP environment itself creates. Publicly available information compounds this exposure. As public agencies, PSAPs are often easy targets for open source intelligence gathering (OSINT): IT and supervisory staff names, computer-aided dispatch (CAD) and other platform vendors, staff email signature formats, and even real-time call activity are accessible through open sources, giving a well-prepared attacker material to make an impersonation convincing before first contact.

Not all awareness training counters this equally. Prümmer et al. (2024), in a systematic review of 142 studies, found that simulation-based and active-learning methods consistently outperform passive delivery, and that effectiveness is tied to job relevance and scenario-based practice. KnowBe4 (2025) reports that roughly 1 in 3 employees will click a phishing link before any awareness training, a rate that drops substantially with tailored training. Reason's (1990) Swiss cheese model of organizational accident causation, widely used across many industries, from medicine to aviation, offers a useful concept: no single defensive layer is perfect, and a breach occurs only when gaps across layers align. This concept maps to the more familiar defense-in-depth, in which technical controls, policies, and human behavioral training each add coverage. So strengthening the human layer, with specific, role-tailored security awareness training, should improve the overall security posture even when other layers remain unchanged, eliminating some of the holes in the Swiss cheese.

Problem Statement

Despite the vulnerabilities we've talked about so far that exist within PSAP systems, most cybersecurity awareness training delivered to PSAP staff is drawn from enterprise frameworks, not designed with the dispatch environment in mind. Annual IT compliance training covers basic phishing, password hygiene, and other baseline concepts. Still, it does not account for the attack vectors we're focusing on that are likely to target PSAPs: social engineering through vendor impersonation, vishing attacks that leverage publicly available organizational data, and spear-phishing emails crafted to mimic the system notifications dispatch staff see every day.

The design premise of this project is that agency staff have meaningful conceptual awareness of cybersecurity threats but lack PSAP-specific scenario exposure to connect that knowledge to their daily work. Generic awareness and job-specific behavioral training are not interchangeable. A staff member may know that phishing exists and still rely on an unreliable verification habit under operational pressure, and may correctly distrust a suspicious caller yet stop short of the verification and reporting steps that would actually resolve the threat. The problem this project addresses is therefore the distance between conceptual security knowledge and reliable behavioral response in the PSAP operating environment.

Results

Project Design and Approach

The project was structured around five phases: stakeholder engagement and scope definition; needs assessment through a pre-training survey; iterative development of training materials; training delivery; and post-training analysis and documentation. Each phase informed

the next, allowing survey findings to directly shape training content before delivery. The project used the pre-training survey for baseline assessment and a matched post-training survey targeting the identified gaps, with the full staff cohort serving as both the training and the measurement population. Initial outreach targeted the agency's Training and Development Manager rather than IT, on the basis that the project solves a training delivery problem, not a technical infrastructure problem. The Training and Development Manager served as the primary point of contact throughout, organized input meetings, facilitated survey distribution, and coordinated logistics with agency leadership and the Information Systems Manager.

Survey Design and Data Collection

A 15-item pre-training survey established a baseline across two dimensions: behavioral knowledge, meaning what staff would actually do given a realistic threat scenario, and self-assessed confidence and training history. The pairing was intentional because self-assessment without behavioral data, or behavioral data without attitudinal context, produces an incomplete picture. Questions 1 through 10 presented scenario-based multiple-choice items drawn from documented PSAP attack vectors, including vendor impersonation calls, CAD system phishing emails, credential theft through fake login pages, suspicious file delivery, and callback number manipulation. Distractors were written to reflect responses that could seem operationally reasonable under time pressure, to detect gaps between conceptual awareness and reliable behavior. Questions 11 through 15 used Likert-style or categorical formats to measure familiarity, perceived threat to PSAPs, self-confidence, recency of prior training, and knowledge of how to report a suspicious call or email. No personally identifiable information was collected, participation was voluntary, and the Google Form used for the survey was configured to disable

email collection and sign-in and to include an explicit anonymity disclosure. The survey was distributed digitally by the Training and Development Manager about one week before training. Of 27 responses received, 1, submitted a week after training was delivered, was excluded from the pre-training baseline; the remaining 26 responses represent a majority of the 29 staff who attended.

Baseline Findings

The 10 behavioral questions produced the results in Table 1. Six questions reached a 100% correct rate, confirming a strong baseline of conceptual awareness across the most recognizable scenarios. Meaningful errors concentrated on Q5 and Q6, the two questions most directly tied to the behavioral gaps the training was designed to address. In aggregate, 9 of 26 respondents (34.6%) missed at least one behavioral question, and Q5 and Q6 together accounted for 7 of those 9.

Table 1*Behavioral Question Results (n = 26)*

Question	Topic	Correct	Error rate
Q1	Suspicious sender address	26/26 (100%)	0%
Q2	Credential request from vendor caller	25/26 (96.2%)	3.8%
Q3	Urgent NCIC lockout email	26/26 (100%)	0%
Q4	Request to run a downloaded file	26/26 (100%)	0%
Q5	Safest way to verify an email link	22/26 (84.6%)	15.4%
Q6	Callback number from unverified caller	23/26 (88.5%)	11.5%
Q7	Primary goal of a phishing email	26/26 (100%)	0%
Q8	OSINT and attacker research awareness	26/26 (100%)	0%
Q9	Response to a convincing fake login page	24/26 (92.3%)	7.7%
Q10	Correct action after suspicious contact	26/26 (100%)	0%

Two gaps drove the training design. On Q5, link verification (84.6% correct), four respondents selected a surface-level verification method, two judging an email by whether it arrived during business hours, and two by whether it looked professional and carried a company logo. Neither approach detects a well-crafted phishing email, which is specifically designed to defeat visual-professionalism checks. The gap matters because it reflects a genuinely held belief: a respondent who trusts a professional-looking email will feel most confident exactly when that confidence is dangerous. On Q6, independent callback verification (88.5% correct): three respondents would refuse to call an attacker-provided number but then offer to continue helping on the current call. These respondents correctly distrusted the provided number but failed to recognize that continued compliance remained a threat. Additionally, refusing the callback without independently verifying or reporting the contact to a supervisor or IT leaves the claimed identity unverified and the potential threat unreported. Two lower-frequency errors were also

noted and addressed in the facilitator script: on Q9 (92.3% correct), two respondents would log into a page reached from an email link, one reasoning that changing to a complex password would secure the account and one trusting a padlock icon, which confirms encryption, not a trustworthy destination; and on Q2 (96.2% correct), one respondent would give a login name to a caller claiming to be a known CAD vendor, the core vulnerability that vendor impersonation exploits.

The attitudinal items appear in Table 2. The most significant finding was the gap between confidence and demonstrated knowledge: 92.3% of respondents described themselves as very or somewhat confident in recognizing an attack, while 34.6% missed at least one behavioral question. This incomplete self-picture is itself a documented vulnerability because attackers rely on targets who believe their awareness is sufficient. At the same time, 92.3% believed the dispatch center is a target for cyberattacks, an attitudinal precondition for engagement with the training, and 73.1% reported cybersecurity training within the past two years. That the Q5 and Q6 gaps persisted even among recently trained staff reinforces the project's central argument that general awareness training does not reliably produce these specific behaviors.

Table 2*Attitudinal and Self-Assessment Results (n = 26)*

Survey item	Response A	Response B	Response C
Q11: Familiarity with phishing / SE	Very familiar: 26.9% (7)	Somewhat familiar: 69.2% (18)	Not at all: 3.8% (1)
Q12: PSAPs are cyberattack targets	Yes: 92.3% (24)	Unsure: 7.7% (2)	No: 0%
Q13: Confidence recognizing attacks	Very confident: 46.2% (12)	Somewhat confident: 46.2% (12)	Not confident: 7.7% (2)
Q14: Received training in past 2 years	Yes: 73.1% (19)	Unsure: 15.4% (4)	No: 11.5% (3)
Q15: Know how to report suspicious email	Yes: 92.3% (24)	Unsure: 7.7% (2)	No: 0%

Training Design in Response to Findings

Survey results were analyzed before training materials were finalized, and the two primary gaps directly shaped all three deliverables. The training presentation is a 15-slide deck covering phishing and vishing fundamentals, PSAP-specific attack vectors, a full scenario walkthrough integrating both threat categories, and behavioral response guidelines, with statistics sourced from primary benchmarking reports. The facilitator script is a trainer-facing document providing narration, discussion prompts, timing guidance, and scenario cues for each slide. The quick-reference card is a single-page handout organized into three panels: phishing red flags, vishing response steps, and a combined verification grid, with agency-specific reporting instructions.

The link-verification gap (Q5) shaped content across all three deliverables. The hover-to-preview technique is taught on a dedicated slide, reinforced in the phishing scenario walkthrough, and printed on the reference card. The facilitator script names the visual-

professionalism misconception directly, allowing the trainer to correct it rather than simply present the correct method. The callback-verification gap (Q6) shaped the core vishing scenario, which models the full correct sequence with scripted dialogue: recognize the red flag, decline the caller-provided number, notify a supervisor, and, if able, independently look up a known organizational number. The scenario discussion includes a caller who pushes back when refused, reflecting the social pressure of a real attempt, and the script draws an explicit distinction between refusing a callback and independently verifying it, so the gap is addressed without implying that the security-minded instinct behind it was wrong.

Two framing decisions also followed from the data. Because the confidence-competence gap showed a cohort that believed it was already competent, each session opened by acknowledging staff's existing awareness and positioning the training as PSAP-specific tools rather than remedial education, keeping it complementary to the agency's existing training. And because the reporting pathway was already well understood (Q10 and Q15), it was briefly reinforced rather than retaught. Discussion of the open-source information an attacker can gather about the dispatch center was delivered verbally using script prompts and deliberately kept off the slides.

Implementation and Delivery

Training was delivered across four instructor-led sessions at the agency on May 19 and 20, 2026, scheduled around shift changes to accommodate the dispatch center's 24-hour staffing without disrupting active dispatch coverage. On the day of delivery, shift supervisory staff requested that each session be reduced from the planned 30 minutes to roughly 15 minutes to accommodate operational scheduling. The instructor adapted by dropping three lower-priority

background slides and trimming inter-slide discussion while preserving the full phishing email walkthrough and the vishing callback script, which carried the behavioral core of the training. This compression is itself a finding: the same operational time pressure that defines the PSAP environment also constrained the delivery of training within it. The facilitator script's timing marks, built for the 30-minute format, allowed the content to be re-paced on the spot, protecting the scenario walkthroughs while trimming lower-priority material.

A total of 29 staff attended, and physical reference cards were distributed to each. Engagement was highest during the vishing scenario, particularly the extended dialogue modeling a caller who applies social pressure while impersonating a known third-party vendor; a scenario that ends after one firm refusal does not prepare staff for what an actual attempt feels like. The instructor's background in public safety and emergency telecommunications provided direct credibility and enabled scenario framing that drew on the operational environment. End-of-session questions clustered around two areas: handling vendor calls that arrive on the main dispatch line rather than the admin line, and whether the existing IT reporting process was the right channel for suspected social engineering.

Outcomes Evaluation

The post-training survey used an identical survey, allowing comparison of the targeted behavioral gaps (Q5, Q6) and the attitudinal measures. It was distributed on May 22, 2026, but an active regional brushfire incident suppressed responses during the initial window; responses arrived between May 26 and 29 as availability normalized, resulting in 13 completed responses. Two constraints shape how the results read. The instrument is anonymous, so the 13 post-training responses cannot be matched to specific pre-training respondents; this is a comparison of

two independent cross-sections of the same small cohort, not a paired measurement. The post-training group is also self-selected and smaller than the baseline (13 versus 26). Results are therefore directional evidence of a shift within the cohort, not a generalizable outcome.

The clearest result is on Q5, the gap that drove the primary design decision: the correct rate rose from 84.6% to 100%, with every post-training respondent selecting hover-to-preview and none selecting the visual-professionalism or business-hours distractors. Q6 did not show the intended improvement, moving from 88.5% to 84.6%. The composition is more informative than the rate: the specific error the training targeted held roughly flat in proportion (11.5% to 7.7%), while a new error absent at baseline appeared, with one respondent choosing to ignore the contact entirely. At a sample of 13, where one respondent is about eight percentage points, this is within sampling noise and does not support a claim of improvement on Q6. The attitudinal measures shifted as intended: very familiar rose from 26.9% to 53.8%; no respondent selected "not at all familiar"; the not-confident responses dropped to zero; and belief that the dispatch center is a target and confidence in reporting both reached 100%. Q14, recent training, moved to 100% but is largely mechanical, since every post-training respondent had just been trained.

Table 3

Pre- and Post-Training Comparison

Measure	Pre-training (n = 26)	Post-training (n = 13)
Q5, link verification	84.6% correct	100% correct
Q6, independent callback	88.5% correct	84.6% correct
Missed at least one behavioral question	34.6%	15.4%

The data supports a narrow but real claim. On the single behavioral gap the training was built to close, the cohort moved from a measurable deficit to a perfect rate, and self-reported

familiarity and confidence rose in parallel. That alignment of a behavioral gain with an attitudinal gain is the most meaningful signal the design can produce at this scale. The unresolved Q6 result and the small, unpaired sample mean the project cannot claim broad behavioral change; what it can claim is that a targeted, needs-assessed intervention moved the specific behavior it targeted.

Limitations

Several limitations constrain these conclusions. The cohort is small and single-site, which limits direct comparison with PSAPs of different sizes or vendor environments. The anonymous instruments were unpaired, preventing individual-level measurement of change, and the post-training sample was smaller and self-selected, both of which reduce statistical power and may bias results toward more engaged staff. The measurement window was short, capturing the immediate post-training response rather than retention over time, and on-site delivery was compressed from 30 minutes to roughly 15 minutes, limiting the depth of discussion and the supporting content each session could cover.

Conclusion

Public Safety Answering Points operate under the exact conditions social engineering is built to exploit, and the awareness training most of them receive is drawn from enterprise-general frameworks that do not account for those conditions or PSAP-specific attack vectors. This project addressed that gap by developing and delivering a needs-assessed, PSAP-specific training program with the agency.

The contribution is both a method and a finding. The method is a replicable sequence: stakeholder engagement through a training rather than IT contact, a paired behavioral and attitudinal needs assessment, survey-driven targeting of specific gaps, and scenario-based delivery supported by a retained reference tool. The specific gaps would differ at another PSAP, or even the same PSAP, but with a different set of staff; the method for finding and targeting those gaps would not. The finding is that a confident, generally trained cohort still carried specific behavioral gaps, and that a targeted intervention closed the primary gap it was designed to address while leaving a second gap unresolved. That distinction is the central point. The distance between conceptual security knowledge and reliable behavioral response under pressure is real and measurable; the Q5 result shows it can be narrowed with operationally specific training, and the unresolved Q6 result shows that narrowing it is not automatic and that some gaps require repeated, deliberate reinforcement.

Three recommendations follow for the agency. First, re-administer the post-training survey once staffing has fully normalized to obtain a larger sample and strengthen the comparison. Second, address the Q6 result directly in the next scenario refresher, since the distinction between declining a suspicious contact and independently verifying it did not fully land. Third, keep the reference card up to date with reporting instructions, since its value depends on the information staying accurate. Periodic scenario refreshers, rather than one-time delivery, are the format most consistent with durable retention. For an environment where a single failure can take a 911 center offline, this argues for treating PSAP-specific training as an ongoing operational practice rather than an annual compliance task.

Reflection

Several lessons emerged from the process itself. Routing the project through the Training and Development Manager rather than IT proved to be the right decision; framing the work as a training problem rather than a technical audit opened up access and cooperation that a security review would not have. The on-site compression from 30 minutes to 15 was the clearest practical lesson, a reminder that operational reality will reshape even a well-prepared plan, and that the timing flexibility built into the facilitator materials in advance is what allowed the adaptation to be smooth rather than disruptive. The Q6 result was a useful corrective to my own expectations. I had assumed that targeted instruction would move both gaps, and learning that it closed one cleanly while leaving the other taught me more about the difficulty of behavioral change than a uniform success would have. Finally, designing from survey data rather than intuition changed not only what the training covered but how each session was framed, and seeing staff engage most with scenarios drawn from their own environment confirmed that relevance, not volume, is what makes this kind of training land.

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